

DL for ENG

LECTURE 11.2

Dynamics, Energy and Efficient Driving

The physics comes back — as ordinary differential equations, not partial ones. Three models say how the car moves, what it cannot do, and what the manoeuvre costs.

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Kinematic bicycle model · DC motor and 18650 pack · INA219 telemetry



What You Will Be Able To Do

THE GOALS, IN PLAIN WORDS

- **write** — the kinematic bicycle model, and name its single parameter
- **calibrate** — a steering map by measurement, not by assumption
- **state** — the grip limit, and why braking and cornering share it
- **choose** — between a soft penalty and a hard layer, and defend the choice
- **explain** — why the energy-optimal speed is not the slowest speed
- **connect** — the pack model here to the one you built in L10.1

WHAT TODAY ASSUMES

from L11.1	the car, and a working policy
from L7.1	hard against soft enforcement
from L10.1	the equivalent-circuit pack model

PHYSICS WITHOUT A PDE

Ordinary differential equations, algebraic constraints, and an objective. The machinery of L7 applies unchanged.

AND ONE UNGLAMOROUS SLIDE

The steering calibration decides whether every later result means anything.

Where the Physics Comes Back

THE CLAIM, STATED PRECISELY

L11.1 had no governing equation. L11.2 has three — kinematics, a friction limit, and an energy model — and none of them is a PDE. They are ordinary differential equations in time, and they enter the loss exactly as a PDE residual did in L7. The machinery is unchanged; only the operator is.

	L 11.1	L 11.2
What constrains the model	labelled images	an ODE residual and a grip limit
What you can verify against	held-out data, then the floor	a measured power trace
What limits you	latency and data	model error and unmodelled friction
Typical failure	the car hits a wall	the car is slow, or flat, or both
The instrument that decides it	a stopwatch	the INA219

Yes, This Is Still a PINN — Mostly

WHERE THE PART 2 MACHINERY APPLIES

- identifying the vehicle parameters is a PINN inverse problem, with ODEs instead of PDEs
- a physics residual in the loss, autograd derivatives, coefficients made trainable
- **optimising the lap is a step beyond** — a non-zero objective, with the physics as a constraint
- so a PINN drives residuals to zero; here you minimise energy subject to them being zero

WHICH PART 1 IDEAS ARE IN USE

from L7.1	the composite residual loss
from L6.1	autograd, Adam, L-BFGS
from L10.2	physics inside an optimisation
from L11.1	the policy being optimised

WHAT IS AND IS NOT A PINN HERE

Identification is one, by any reasonable definition. The lap optimisation is optimal control through a differentiable model, which L10.2 also did.

AND THAT IS THE POINT

A differentiable model is one you can optimise through. Slide 14 is why that matters.

The Kinematic Bicycle Model

FOUR STATES, TWO INPUTS

$$\dot{x} = v \cos \theta, \quad \dot{y} = v \sin \theta, \quad \dot{\theta} = \frac{v}{L} \tan \delta, \quad \dot{v} = a$$

- collapse the two front wheels onto one, the two rear onto one
- state is position, heading and speed; inputs are steering angle and acceleration
- the only vehicle parameter is the wheelbase
- mass, tyre stiffness and weight transfer are all absent

THE ONE LINE TO REMEMBER

A steering command buys curvature, not yaw rate. Everything on slides 6, 15 and 17 follows from that sentence.

CURVATURE AND TURN RADIUS

$$\kappa = \frac{\tan \delta}{L} \quad R = \frac{L}{\tan \delta} \quad R_{\min} = \frac{L}{\tan \delta_{\max}}$$

With $L \approx 0.17$ m and full lock near 30° , the tightest circle the car can describe is about 0.29 m in radius — roughly one and a half car lengths.

THE SLIP-ANGLE REFINEMENT

$$\beta = \arctan \left(\frac{l_r \tan \delta}{L} \right)$$

Referencing to the centre of mass instead of the rear axle introduces a body slip angle β . It is a genuine improvement and it is not worth it here: at JetRacer speeds the correction is smaller than your measurement error in L .

What the Model Assumes, and When It Breaks

No lateral slip

assumes

Each wheel rolls in the direction it points. The tyre generates whatever side force is required, instantly and for free.

Holds while lateral acceleration stays well below the friction limit — which is exactly the regime slide 6 defines.

No weight transfer

assumes

Load stays evenly distributed under braking and cornering, so grip is constant at all four contact patches.

With real springs and a high pack this car rolls and pitches visibly. It shows up as understeer on turn-in, and the model will not predict it.

Instant steering

assumes

The servo reaches the commanded angle immediately; there is no actuator lag and no rate limit.

The servo needs of order 100 ms for a large step. At a 20 Hz loop that is two commands, and it matters when you fit parameters.

Planar and rigid

assumes

No suspension travel, no ground undulation, no chassis flex, and a perfectly level floor.

Knowingly false: the Pro has adjustable oil-filled shocks and independent suspension on all four corners. Ignored, not absent.

Speed is commanded

assumes

Longitudinal acceleration a is treated as a free input the drivetrain can supply on demand.

This is the assumption L11.2 removes. Slides 8 to 10 replace a with a motor, a pack, and a torque the car actually has.

The kinematic model is not an approximation to a better model you will later write. It is the correct model for this vehicle in this speed range, and the fifth row is the only assumption L11.2 asks you to discard.

From Command to Steering Angle

- the model takes an angle in radians; the car takes a normalised servo command
- the map between them is not linear, because linkages have geometry
- **so measure it** — drive steady circles at fixed commands and invert
- there is also an offset: a car that drifts on zero command has a trim error
- fit a polynomial or a lookup table, not a neural network

Twenty minutes of calibration at the start of the session. Every team that skips it spends longer than that later, arguing with a fit that will not converge.

THE CALIBRATION DRIVE

- | | | |
|---|--------------------------|--|
| 1 | Fix throttle low | constant, slow, well below the grip limit |
| 2 | Fix the steering command | one value, held for a full circle |
| 3 | Measure the circle | chalk the floor, or time a full revolution |
| 4 | Invert the geometry | $\delta = \arctan(L / R)$ for that command |
| 5 | Repeat, both directions | five commands each way; the car is not symmetric |

WHY THIS SLIDE EXISTS

Every later result depends on it. A model fitted through an uncalibrated steering map will identify a wrong wheelbase, a wrong friction coefficient, and a plausible-looking energy curve that is wrong by a constant factor.

What the Car Physically Cannot Do

THE CORNERING LIMIT

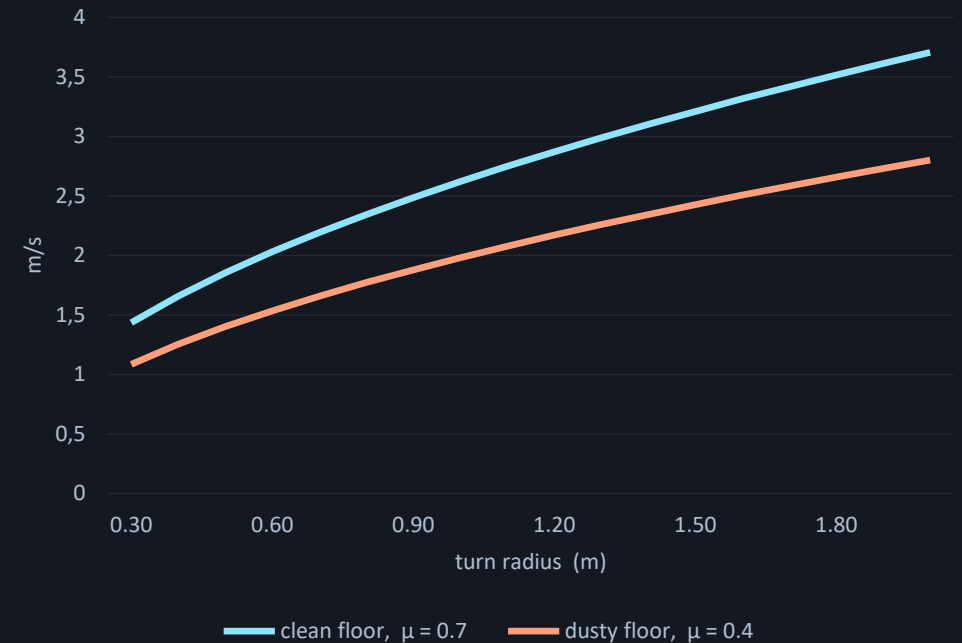
$$a_y = \frac{v^2}{R} \leq \mu g \quad \Rightarrow \quad v_{\max}(R) = \sqrt{\mu g R}$$

- lateral acceleration in a steady turn is v^2 over R
- the contact patches can supply no more than μg
- μ is a property of your floor and your tyres together, and it changes
- **combined loading is stricter** — braking and cornering share one friction

$$\sqrt{a_x^2 + a_y^2} \leq \mu g$$

measure μ once per session — it is not a constant of nature

MAXIMUM SPEED AGAINST TURN RADIUS



A 0.5 m corner caps the car at 1.9 m/s on a clean floor and 1.4 m/s on a dusty one. L11.1 assumed 2 m/s throughout.

Two Ways to Enforce a Constraint

YOU HAVE SEEN THIS BEFORE

This is the soft-versus-hard distinction from L7.1, transplanted. There it was a boundary condition; here it is a grip limit. The argument for each is identical, and so is the failure mode of choosing wrongly.

SOFT — A PENALTY IN THE LOSS

$$\mathcal{L}_{\text{con}} = \frac{1}{N} \sum_k [\max(0, a_{y,k} - \mu g)]^2$$

- a soft penalty is zero inside the feasible set and quadratic outside
- **it is a preference, not a guarantee** — the optimiser will trade a small violation
- a hard constraint squashes the output through a bounded function

HARD — A LAYER THE POLICY CANNOT PASS

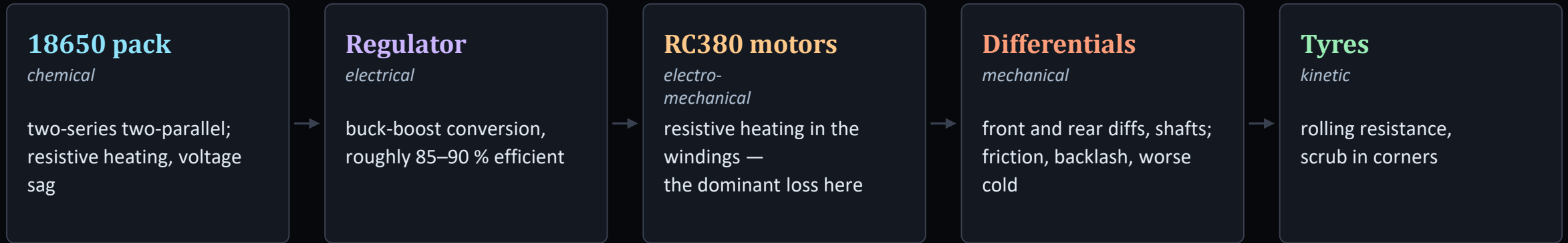
$$\delta = \delta_{\max} \tanh(\hat{\delta}), \quad v = v_{\max}(R) \sigma(\hat{v})$$

Squash the raw network output through a bounded function scaled by the limit. The steering command cannot exceed full lock and the speed command.

Enforce the grip limit hard and the smoothness preference soft. A car that is merely discouraged from exceeding its tyres will exceed them.

Where the Joules Actually Go

FROM CELL TO CONTACT PATCH — EVERY STAGE TAKES A CUT



The Nano is a stage too — and it never stops drawing.

- in 5 W mode the compute board, camera and display draw a few watts continuously
- whether the car is moving or not
- aerodynamic drag, by contrast, is negligible at 2 m/s
- both facts are the reverse of a full-size electric vehicle

THE INVERTED INTUITION

In a full-size electric vehicle, aerodynamic drag dominates at speed and the auxiliary load is a rounding error. On a JetRacer both facts reverse: drag is negligible, the constant compute load is not, and the largest single loss is resistive heating in the motor windings. Carrying the road-vehicle intuition across will give you the wrong optimal speed.

The Brushed DC Motor, in Three Equations

TORQUE, BACK-EMF, AND THE VOLTAGE BALANCE

$$\tau = k_t i, \quad e = k_e \omega, \quad V = iR_a + k_e \omega$$

- torque is proportional to current; back-EMF is proportional to speed
- the applied voltage splits between resistance and back-EMF

THE POWER SPLIT

$$P_{\text{elec}} = Vi = \underbrace{i^2 R_a}_{\text{heat}} + \underbrace{\tau \omega}_{\text{mechanical}}$$

- some of the input power turns the wheels; the rest becomes heat
- copper loss is quadratic in current, so quadratic in torque

LONGITUDINAL DYNAMICS

$$m\dot{v} = \frac{G k_t i}{r_w} - C_{rr}mg - \frac{1}{2}\rho C_d A v^2$$

Gear ratio and wheel radius turn motor torque into tractive force, and enter only as the lumped constant above. The last term is the aerodynamic one, and on this car you may keep it only to demonstrate that it does not matter.

RC380 BRUSHED, AND NO ENCODER ON IT

no encoder fitted	the Pro has none — speed must come from lap timing or from vision
traction constant	torque constant, final drive and wheel radius, only ever as one product
armature resistance	winding, brush and wiring together; it rises as the motor warms
rolling coefficient	tyre and floor together, and it absorbs differential drag too

The Pack: L10.1, on Hardware You Can Hold

EQUIVALENT-CIRCUIT MODEL AND STATE OF CHARGE

$$V_{\text{term}} = \text{OCV}(z) - iR_{\text{int}}, \quad \dot{z} = -\frac{i}{Q}$$

- **the simplest useful model** — an open-circuit voltage depending on state of charge
- in series with an internal resistance
- state of charge follows by coulomb counting
- an RC branch captures the relaxation transient, and is the first refinement worth adding
- this is L10.1's model, on hardware you can hold

WHY SAG IS A CONTROL PROBLEM, NOT A NUISANCE

Internal resistance is what connects L11.2 to the 5 W rule from L11.1. A hard acceleration pulls the terminal voltage down; if the compute rail is already near its limit, the Nano browns out and resets mid-run. The energy model and the reset you saw last week are the same equation.

THE PACK, SETTLED

Four cells, 8.4 V, two in series and two in parallel.

The series pair sets the voltage; the parallel pair doubles the capacity and roughly halves the internal resistance. Both matter — the pack sags less under hard acceleration than a bare two-cell string would, and it carries twice the charge at the same voltage.

Cells must be under 67 mm, so protected cells with a plate do not fit. Measure the resting voltage anyway: a healthy charged 2S pack sits near 8.4 V, and well below that you are modelling a tired pack, which is a different pack.

The INA219: What It Actually Cost

A SHUNT, AN AMPLIFIER, AND TWO VOLTAGES

$$i = \frac{V_{\text{shunt}}}{R_{\text{shunt}}}, \quad P = V_{\text{bus}} i$$

A small resistor sits in the supply line. The chip measures the voltage across it and the bus voltage on one side of it, and current follows from Ohm's law. Multiply and you have instantaneous power, over I²C, from the car itself.

This is the component that makes the MiniProject an experiment. Without it you would be comparing a predicted energy against another prediction; with it, the model has something to be wrong about.

Configure it deliberately. The chip offers a choice of conversion time and on-chip averaging, and the defaults are chosen for slowly varying loads rather than for a PWM drive on a moving vehicle.

Read the whole car, not one branch. If the shunt sits downstream of the compute rail you will measure traction only, and the hotel load — the term that decides the optimum — is invisible to you.

Configure the chip once, at the top of the notebook, and record the configuration in your log header. A trace whose sampling regime is unknown cannot be compared against anyone else's.

THE ALIASING TRAP

The motor drive is switched at a few kilohertz. Sampling the current at 20 Hz with a short conversion window does not measure the average of that ripple — it measures wherever in the PWM cycle each sample happened to land, and the resulting trace is noise dressed as data.

Set the conversion time and averaging so that each reported sample integrates over many switching periods. You want the chip to do the averaging, in hardware, before the value reaches you.

WHAT TO RECORD, EVERY SAMPLE

<i>t</i>	monotonic clock, not wall time
<i>V, i, P</i>	bus voltage, current, product
<i>throttle,</i> <i>steering</i>	the commands that were active
<i>v</i>	measured speed, if you have it

Turning Samples into an Energy Figure

INTEGRATE POWER OVER THE RUN

$$E = \int_{0^T} Vi \, dt \approx \sum_k \frac{1}{2} (P_k + P_{k+1}) \Delta t_k$$

Trapezoidal integration over irregular timestamps, which is what you will actually have. Do not assume a fixed Δt : the loop from L11.1 jitters, and treating a 90 ms interval as 50 ms understates the energy by nearly half over that step.

Log from the control loop, not from a separate thread with its own timer. You need the current sample and the command that produced it to carry the same timestamp, or the identification on slide 13 has nothing to align.

Logging costs loop rate. Buffer in memory and write once at the end of the run — an fsync inside the loop is exactly the blocking call L11.1 warned about, and it will cost you frames.

Report energy per lap and mean power separately. They answer different questions, and a team that quotes only one of them cannot tell a fast efficient lap from a slow wasteful one.

THE LOOP, EXTENDED FROM L11.1

- 1 **Capture** camera frame
- 2 **Infer** CNN forward pass
- 3 **Constrain** grip limit, hard layer
- 4 **Actuate** throttle and steering
- 5 **Sample** INA219: V, i, t
- 6 **Buffer** append, never write

Steps 3, 5 and 6 are what L11.2 adds. Steps 5 and 6 must be cheap, or they will quietly undo the frame rate you fought for last week.

Fitting the Model to a Drive

THIS IS AN INVERSE PROBLEM, AND YOU HAVE SOLVED ONE BEFORE

The structure of the model is known and four of its coefficients are not. That is exactly the inverse setting from L7: unknown parameters promoted to trainable variables, recovered by minimising a residual against measured data. The only difference is that the data came off a vehicle rather than a solver.

$\lambda = \{C_{rr}, Gk_t/r_w, R_a, R_{int}\}$ trainable

Drive a rich excitation, not a demonstration lap. Steps, ramps, a coast-down and a steady cruise — a policy that only ever cruises leaves armature resistance and rolling resistance unidentifiable, because nothing in the data distinguishes them.

The coast-down is the single most informative manoeuvre available to you. Cut the drive and log the deceleration: with zero current the electrical terms vanish and the rolling coefficient falls out of a two-parameter fit almost by itself.

Scale the parameters before optimising. Resistance is of order one ohm and capacity of order ten thousand coulombs, and an optimiser handed both at once will make no progress on one of them.

Report the identified values with units and an honest uncertainty. A parameter quoted to four significant figures from a single drive is a claim you cannot support.

HOW TO KNOW WHETHER YOU BELIEVE THE FIT

Hold out a run

Fit on one drive, predict another. A model that only reproduces its own training run has fitted the drive, not the car.

Check the signs

A negative rolling coefficient or a negative resistance is not a small numerical issue. It means the model is absorbing something you have not included.

Compare magnitudes

Resistance of a few ohms, rolling coefficient of a few hundredths. Values orders away from these indicate a units error, usually in the gear ratio.

Refit warm and cold

Winding resistance rises appreciably as the motors heat. If the two fits differ, that is physics, and it belongs in your report.

The Loss, Assembled

THREE TERMS, ONE OBJECTIVE

$$\mathcal{L}(\theta, \lambda) = \mathcal{L}_{\text{data}} + \lambda_{\text{dyn}} \mathcal{L}_{\text{dyn}} + \lambda_{\text{con}} \mathcal{L}_{\text{con}}$$

$$\mathcal{L}_{\text{dyn}} = \frac{1}{N} \sum_{k=1}^N \|\dot{\mathbf{x}}(t_k) - \mathbf{f}(\mathbf{x}(t_k), \mathbf{u}(t_k); \lambda)\|^2$$

The data term anchors the trajectory to what the car did. The dynamics term penalises departure from the model of slides 3 and 9. The constraint term handles whatever slide 7 did not enforce by construction.

The derivative in the residual is taken by automatic differentiation with respect to time — one input, not several. This is the whole of the difference from L7, and it makes the problem cheaper rather than harder.

Weighting is the same difficulty it always was. Non-dimensionalise each residual before choosing λ , or the weights end up encoding unit conversions instead of relative importance.

Everything above is L7 with one input instead of several. If that sentence is not obvious by now, revisit L7.1 before the lab rather than during it.

WHAT CHANGED FROM L7

Operator

$\partial/\partial t$ only

a system of ODEs, not a PDE

Collocation

along a trajectory

no interior sampling of a domain

Boundary data

an initial condition

one state, at $t = 0$

Unknowns

θ and λ jointly

network weights and physical coefficients

Ground truth

an instrument

a measured power trace, not a manufactured solution

The Layer the Policy Cannot Override

The perception network from L11.1 proposes. It does not dispose. Between its output and the actuators sits a filter that knows the wheelbase, the friction limit and the current speed, and that clamps any command the vehicle cannot execute.

This is not a safety afterthought bolted onto a learned controller. It is the hard-enforcement construction of slide 7, placed in the loop rather than in the loss, and it is active during training as well as deployment.

The filter must be cheap. It is a handful of arithmetic operations on scalars, evaluated once per frame — if it ever appears in your loop-rate profile, it has been implemented wrongly.

Give it authority over the throttle too. Steering saturation alone does not prevent a car from arriving at a corner too fast to take it; the speed ceiling is the constraint that actually keeps the car on the floor.

And keep the L11.1 failsafe underneath everything. A late frame, a stale command or a malformed reading still means stop — a constrained policy acting on old information is constrained to do the wrong thing accurately.

Write the filter before you write the policy. It is the one component in this exercise that must be correct on the first run rather than the tenth.

COMMAND PATH, PROPOSAL TO ACTUATION

CNN output raw steering and throttle proposal

Calibration map command $\rightarrow \delta$ in radians

Curvature $\kappa = \tan \delta / L$, hence R

Grip ceiling $v \leq v(\mu g R)$, evaluated now

Clamp the binding limit wins

Actuate servo and motor commands

Rows four and five are the whole of the safety argument, and together they are about six lines of code.

Asking for the Cheapest Lap

ENERGY AND TIME IN ONE OBJECTIVE

$$J = \int_{0^T} [P(t) + w]dt = E + wT$$

Minimise energy alone and the answer is trivial and useless: never move. Minimise time alone and you have the L11.1 objective with a flat battery. The weight w prices a second in joules and is the entire design decision on this slide.

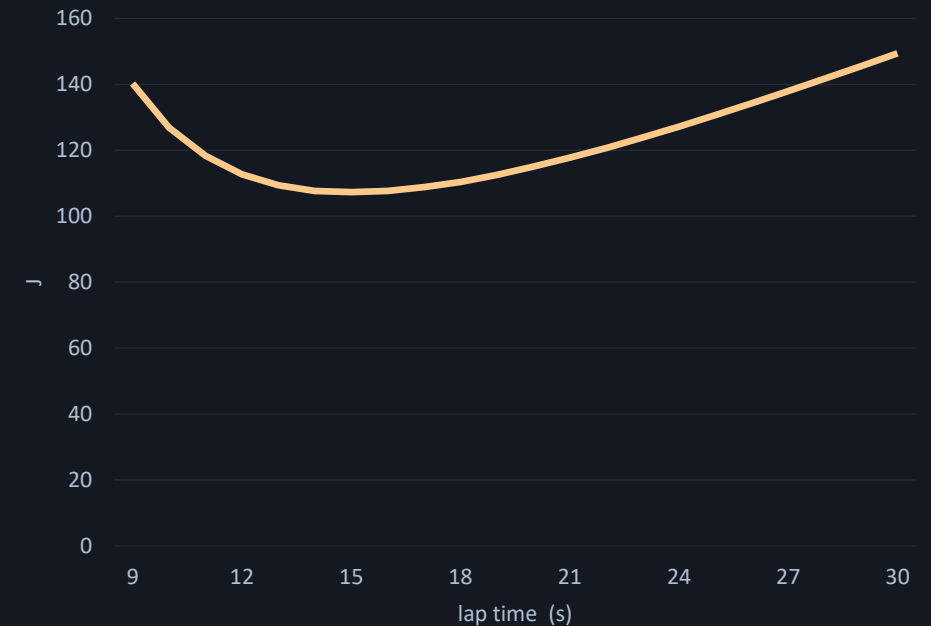
Subject to everything already established — the kinematics of slide 3, the grip limit of slide 6, the drivetrain of slides 8 and 9, and the pack of slide 10. The optimisation is only as trustworthy as the parameters identified on slide 13.

Solve it directly. Discretise the lap into segments, treat the speed in each as a variable, and hand the constrained problem to any nonlinear programme. A learned policy is a fine alternative and is not required to see the physics.

STATE w EXPLICITLY IN YOUR REPORT

Two teams reporting different optimal speeds have usually not disagreed about physics — they have chosen different w and not said so. A Pareto curve is the honest deliverable; a single optimum is a curve evaluated at an undisclosed point.

ENERGY PER LAP AGAINST LAP TIME



Illustrative, with a 4.5 W hotel load over a 20 m lap. The curve has an interior minimum — and it is not at the slow end.

Why the Efficient Lap Is the Smooth One

THE CONVEXITY ARGUMENT

$$\overline{i^2} \geq \bar{i}^2 \quad \Rightarrow \quad \text{smooth effort costs less heat}$$

Winding loss is quadratic in current, and current is proportional to tractive force. For a fixed mean force, the mean square is smallest when the force is constant — that is Jensen's inequality, not a heuristic.

So a lap driven at steady effort dissipates less heat than a lap of equal average effort driven in bursts. The penalty for surging is paid twice: once in the acceleration, once in the braking that follows.

AND THERE IS NO REGENERATION

$$E_{\text{lost}} = \frac{1}{2} m (v_{\text{in}}^2 - v_{\text{apex}}^2)$$

The kinetic energy carried into a corner and shed before the apex does not return to the pack. It becomes heat in the windings and the tyres, and every joule of it was bought at the pack's efficiency on the way in.

Which gives the practical rule directly: arrive at the corner already at the speed the grip limit permits. Braking late is fast in racing, where energy is free. Here it is not.

THE RESULT THE EXERCISE IS BUILT AROUND

The energy-optimal lap is not the slow lap. It is the lap with the flattest speed profile that still respects the grip limit at every corner — fast on the straights only to the extent that the following corner allows, and never faster. This is a physical prediction, it is counter-intuitive to most students, and the INA219 will settle it in one afternoon.

Naïve fast

full throttle, brake at the corner — shortest time, highest energy

Naïve slow

constant low speed throughout — lowest peak power, worst energy per lap

Grip-limited smooth

speed set by upcoming curvature — near-minimum energy, close to fastest

How These Models Fail

μ was measured yesterday

Grip was identified on a clean floor and the demonstration runs after six other teams have shed dust and tyre rubber.

Symptom: the car understeers wide at speeds that worked. Fix: re-measure at the start of every session.

The pack was not full

Energy figures compared across runs at different states of charge. Internal resistance and terminal voltage both depend on it.

Fix: log resting open-circuit voltage before each run and report it alongside the energy.

Parameters absorbed an error

An uncalibrated steering map or a wrong gear ratio does not show up as a bad fit. The optimiser hides it in the rolling coefficient.

Symptom: an excellent fit with a physically absurd coefficient. Fix: check magnitudes, not residuals.

Warm motors, cold model

Winding resistance rises with temperature, so a model fitted on the first lap over-predicts efficiency on the tenth.

Fix: fit on a warm car, or let armature resistance depend on accumulated energy and identify that too.

The optimum was never tested

A speed profile computed from an identified model, reported as the answer, and never driven on the floor.

Fix: drive it and measure it. The whole point of the INA219 is that this claim is checkable.

The first and third catch everyone. Neither is a modelling error — one is a measurement discipline failure and the other is an identifiability failure, and no amount of training fixes either.

Ex_11.2 — Identify, Constrain, Optimise

TASKS

- 1 Measure L and calibrate the steering map over five commands in each direction.
- 2 Instrument the loop: INA219 sampling, buffered logging, energy by trapezoidal integration.
- 3 Drive an excitation run — steps, ramps, a coast-down, a steady cruise — and log it.
- 4 Identify the rolling, traction, armature and internal-resistance parameters, then validate on a held-out run.
- 5 Add the grip-limit clamp as a hard layer beneath the L11.1 policy and confirm it binds.
- 6 Compute an energy-optimal speed profile, drive it, and compare predicted with measured energy.

QUESTIONS — AND WHERE THIS DECK ANSWERS THEM

- | | |
|---|-----------|
| Why does curvature, not yaw rate, follow from a steering command? | sl. 3 |
| Why is a small excursion past the grip limit not an acceptable trade? | sl. 7 |
| Why can a slower lap consume more energy than a faster one? | sl. 8, 16 |
| Why can the torque constant and gear ratio not be separated? | sl. 9 |
| Why is the coast-down worth more than a demonstration lap? | sl. 13 |
| Why is the smooth profile cheaper at equal average effort? | sl. 17 |

Runs on the same course as Ex_11.1, with the same car and the same policy. What changes is that the car is now instrumented and the claims are now checkable.

The Efficient Lap

THE BRIEF

Complete a fixed course without collision, minimising energy per lap at a lap time you declare in advance. You choose w , you state it, and you defend it. Report predicted energy from your identified model alongside measured energy from the INA219, and account for the gap.

Completes the course

No collision, no intervention, no stall. A run that does not finish does not score, whatever its power trace shows.

Predicted vs measured

The headline number. Not the size of the error — the quality of the explanation for it.

Constraint held

Evidence that the grip limit bound at least once and was never violated. A log excerpt, not an assertion.

Pareto, not a point

At least three (time, energy) pairs from real runs, with w stated for each.

Reproducible

Pack state, floor condition, μ , and identified parameters recorded. Another team should be able to repeat it.

A team that predicts 118 J, measures 141 J, and explains the difference as warm windings has done better physics than a team that predicts 118 J, measures 119 J, and cannot say why.

Safety carries over unchanged from L11.1: one driver, one observer with a hand on the stop, throttle capped in software until the loop is measured, and the battery rules are not negotiable.

Key Takeaways

- 1 Physics-informed does not mean PDE**

Three ODEs — kinematics, grip, drivetrain — enter the loss exactly as a PDE residual did in L7. The operator changed; the machinery did not.
- 2 Enforce grip hard, not softly**

A penalty is a preference the optimiser will trade away. A bounded layer scaled by $\sqrt{\mu g R}$ cannot be traded, and on this vehicle the difference is a crash.
- 3 Identify from excitation, not demonstration**

Steps, ramps and above all a coast-down. A cruise leaves armature and rolling resistance indistinguishable, and the fit will look excellent anyway.
- 4 The instrument is the point**

The INA219 turns the MiniProject from a simulation into an experiment. For the first time in this course, the model can be wrong about something it cannot argue with.
- 5 The cheapest lap is smooth, not slow**

Quadratic winding loss plus a constant compute load put the energy optimum in the interior. Flatten the speed profile against the grip limit and stop braking energy you paid for.

The car is still the benchmark. It now reports what it cost.

Where to Look

Kong, Pfeiffer, Schildbach & Borrelli (2015)

Kinematic and dynamic vehicle models for autonomous driving control design. The standard reference for when the kinematic model is adequate and when it is not — read it for the validity argument, not the equations.

Plett, Battery Management Systems, Vol. 1

Equivalent-circuit models, coulomb counting, and parameter identification. The source behind L10.1, and the place to go when your identified internal resistance misbehaves.

Raissi, Perdikaris & Karniadakis (2019)

The PINN paper from L7. Section 4 is the inverse problem, which is exactly what slide 13 asks you to solve — with an ODE where they used a PDE.

Rajamani, Vehicle Dynamics and Control

The textbook treatment: bicycle models, tyre forces, the friction circle. Chapters 2 and 13 cover everything used in this lecture, at more depth than the exercise needs.

INA219 datasheet, Texas Instruments

Read the conversion-time and averaging registers specifically. The aliasing failure on slide 11 is entirely avoidable from this document.

Waveshare JetRacer Pro AI Kit wiki

Still the authority on the hardware, and still the place to check the pack configuration before you trust any energy figure in this deck.

The vendor documentation remains the authority on the hardware; the academic references are the authority on the models. Where they disagree about this car, the hardware wins.

Questions

TEN TO TEST WHETHER TODAY LANDED

- what is the only vehicle parameter in the bicycle model?
- why is the steering map not linear?
- why do braking and cornering share one budget?
- when is a soft penalty adequate, and when is it not?
- why is drag negligible here but not on a road car?
- why does an interior energy-optimal speed exist at all?

AND FOUR MORE

seven

why is inductance omitted from the motor model?

eight

what does coulomb counting drift on, and why?

nine

what does an RC branch add to the pack model?

ten

which single measurement decides whether the rest is meaningful?

THE ONE WITH NO SETTLED ANSWER

Your six cars are nominally identical. Should they share one model?

BEFORE L12.1

Ex_11.2. Bring your steering calibration curve.

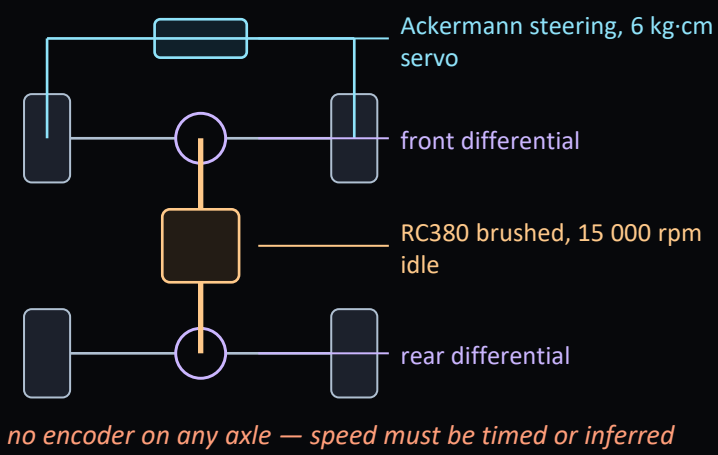
Energy Optimisation of Navigation with the JetRacer

THE CAR AS DELIVERED



- 4WD**
all four wheels driven
- differentials**
front and rear, shaft-linked
- 8.4 V**
four 18650s, 2S2P
- no encoder**
nothing measures speed

DRIVELINE



WHAT MATTERS FOR ENERGY

Compute	Jetson Nano 4 GB, 5 W mode
Pack	4 × 18650, 8.4 V, 2S2P
Motors	RC380 brushed, 15 000 rpm
Transmission	front and rear differentials
Power monitor	INA219 over I ² C, bus + shunt
Hotel load	a few watts, always on
Speed sensing	none fitted — no encoder

Wheelbase measure it yourself

Waveshare, Pro column.

WHAT YOU BUILD ON TOP OF IT

- 1 **Instrument**
INA219 in the loop, buffered log, trapezoidal energy
- 2 **Identify**
coast-down and excitation — no encoder, so time it
- 3 **Constrain**
grip-limit clamp as a hard layer, not a penalty
- 4 **Optimise**
speed profile minimising E + wT, then drive it

WHERE YOU START FROM

Your own Ex_11.1 policy, driving the course at one fixed throttle.

It has no idea what it costs. It cannot slow for a corner, it cannot spend less on a straight, and it treats the pack as infinite.

Nothing in it is wrong. It is simply optimising one axis of a two-axis problem.

THE ONE NUMBER THAT DECIDES THE EXERCISE

Predicted energy against measured energy, on a lap you computed before you drove it.

Everything else — the identification, the clamp, the profile — exists to make that comparison mean something. A team whose prediction is close for a reason they can state has done the exercise. A team whose prediction is close by luck has not.

The Efficiency Trophy — Six Cars, Two Leaderboards

THE WEIGHT IS FIXED, AND ANNOUNCED BEFORE YOU START

Every team optimises the same cost functional with $w = 5 \text{ J/s}$ — one second of lap time is priced at five joules. You may not choose your own w for the scored run. Choosing w is the design decision of slide 16; fixing it is what makes six teams comparable.

BOARD A — EFFICIENCY

$$\eta = \frac{J_0}{J}, \quad J = E_{\text{lap}} + w T_{\text{lap}}$$

J is the scored lap's cost; J_0 is the same car's baseline, driven by the stock fixed-throttle policy before the session. Higher η is better, and $\eta > 1$ means you beat the car you were given.

Normalising per car is deliberate: the six cars differ in pack health, tyre wear and gearbox friction, and an unnormalised board would rank the hardware, not the teams.

GATES — FAIL ANY ONE AND THE RUN DOES NOT SCORE

- **Completes** three consecutive laps, no contact, no intervention
- **Clamp binds** log shows the grip limit binding at least once
- **Clamp holds** log shows lateral acceleration never exceeding μg
- **Committed** predicted energy timestamped before the run

BOARD B — MODEL FIDELITY

$$\phi = \frac{E_{\text{meas}}}{E_{\text{meas}} + |E_{\text{pred}} - E_{\text{meas}}|}$$

Predicted energy comes from your identified model and your computed speed profile, and must be committed in writing before the run. Measured energy comes from the INA219. $\phi = 1$ is a perfect prediction.

This board is why the exercise is physics rather than tuning. A team can win Board A by driving well and still finish last here.

THE TROPHY

Champion is the lowest rank sum across both boards. A tie is broken in favour of Board B — on this course we would rather reward the team that understood its car than the team that drove it hardest.